KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3081 **3 Credits: 3** **Program Core** **Source-grounded**

Digital Circuits and Systems

 To provide a thorough knowledge in number system, sequential and

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3081 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3081.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Instrumentation Engineering
Course code	3081
Course title	Digital Circuits and Systems
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L:2 T:1 P:0) Periods per semester: 45
Periods per semester	45
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

21 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide a thorough knowledge in number system, sequential and
- combinational circuits.
- To develop digital circuits to solve real world problems.

Course outcomes

CO	Official outcome	Study level
CO1	11 Applying Boolean expressions Design the combinational circuits such as full	See official syllabus
CO2	adder, full subtractor, Multiplexer, gray code 13 Applying converter Design the sequential logic circuits such as	See official syllabus
CO3	14 Applying asynchronous and synchronous counter	See official syllabus
CO4	Describe the working of ADC and DAC 5 Understanding	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Apply the K map and Logic rules to simplify the Boolean expressions	3	As specified
Module 2	Multiplexer, Gray code converter	6	As specified
Module 3	Synchronous counter	8	As specified
Module 4	Describe the working of ADC and DAC	4	As specified

MODULE 1

Apply the K map and Logic rules to simplify the Boolean expressions

This module is presented from the official syllabus scope for Digital Circuits and Systems. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Apply the K map and Logic rules to simplify the Boolean expressions
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 4 Applying operations on different number systems

4 Applying operations on different number systems is an official topic in Module 1 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying operations on different number systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying operations on different number systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying operations on different number systems means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Applying operations on different number systems □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Build logical expressions using logic gates 3 Applying Simplify the Boolean expressions using K map and

Build logical expressions using logic gates 3 Applying Simplify the Boolean expressions using K map and is an official topic in Module 1 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Build logical expressions using logic gates 3 Applying Simplify the Boolean expressions using K map and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, build logical expressions using logic gates 3 applying simplify the boolean expressions using k map and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Build logical expressions using logic gates 3 Applying Simplify the Boolean expressions using K map and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Build logical expressions using logic gates 3 Applying Simplify the Boolean expressions using K map and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▾ 4 Applying logic rules Number systems - Decimal, Binary and Hexa Decimal number systems - conversion - use of binary codes - types of binary codes - Binary Coded Decimal, Excess 3 code, Gray code, ASCII code Binary addition, subtraction, multiplication and division - 1’s complement and 2’s complement subtraction - introduction to logic gates - AND, OR, NOT, NAND, NOR, EX-OR and EX-NOR operations - universal property of NAND and NOR gates - realization of AND, OR, NOT, EX-OR and EX-NOR -Laws of Boolean algebra and De- Morgan's theorems - Sum Of Products (SOP) expression, Product Of Sum (POS) expression - min term and max term - simplification of Boolean expressions using logic rules - truth tables - Karnaugh map - 2, 3, 4 variables - K-map reduction - don't cares in K- map-advantages and disadvantages of K- Map- exercise problems Design the Combinational circuits such as full adder, full subtractor,

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Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying logic rules Number systems - Decimal, Binary and Hexa Decimal number systems - conversion - use of binary codes - types of binary codes - Binary Coded Decimal, Excess 3 code, Gray code, ASCII code Binary addition, subtraction, multiplication and division - 1’s complement and 2’s complement subtraction - introduction to logic gates - AND, OR, NOT, NAND, NOR, EX-OR and EX-NOR operations - universal property of NAND and NOR gates - realization of AND, OR, NOT, EX-OR and EX-NOR -Laws of Boolean algebra and De- Morgan's theorems - Sum Of Products (SOP) expression, Product Of Sum (POS) expression - min term and max term - simplification of Boolean expressions using logic rules - truth tables - Karnaugh map - 2, 3, 4 variables - K-map reduction - don't cares in K- map-advantages and disadvantages of K- Map- exercise problems Design the Combinational circuits such as full adder, full subtractor, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying logic rules number systems - decimal, binary and hexa decimal number systems - conversion - use of binary codes - types of binary codes - binary coded decimal, excess 3 code, gray code, ascii code binary addition, subtraction, multiplication and division - 1’s complement and 2’s complement subtraction - introduction to logic gates - and, or, not, nand, nor, ex-or and ex-nor operations - universal property of nand and nor gates - realization of and, or, not, ex-or and ex-nor -laws of boolean algebra and de- morgan's theorems - sum of products (sop) expression, product of sum (pos) expression - min term and max term - simplification of boolean expressions using logic rules - truth tables - karnaugh map - 2, 3, 4 variables - k-map reduction - don't cares in k- map-advantages and disadvantages of k- map- exercise problems design the combinational circuits such as full adder, full subtractor, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying logic rules Number systems - Decimal, Binary and Hexa Decimal number systems - conversion - use of binary codes - types of binary codes - Binary Coded Decimal, Excess 3 code, Gray code, ASCII code Binary addition, subtraction, multiplication and division - 1's complement and 2's complement subtraction - introduction to logic gates - AND, OR, NOT, NAND, NOR, EX-OR and EX-NOR operations - universal property of NAND and NOR gates - realization of AND, OR, NOT, EX-OR and EX-NOR -Laws of Boolean algebra and De- Morgan's theorems - Sum Of Products (SOP) expression, Product Of Sum (POS) expression - min term and max term - simplification of Boolean expressions using logic rules - truth tables - Karnaugh map - 2, 3, 4 variables - K-map reduction - don't cares in K- map-advantages and disadvantages of K- Map- exercise problems Design the Combinational circuits such as full adder, full subtractor, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 4 Applying logic rules Number systems - Decimal, Binary and Hexa Decimal number systems - conversion - use of binary codes - types of binary codes - Binary Coded Decimal, Excess 3 code, Gray code, ASCII code Binary addition, subtraction, multiplication and division - 1’s complement and 2’s complement subtraction - introduction to logic gates - AND, OR, NOT, NAND, NOR, EX-OR and EX-NOR operations - universal property of NAND and NOR gates - realization of AND, OR, NOT, EX-OR and EX-NOR -Laws of Boolean algebra and De- Morgan's theorems - Sum Of Products (SOP) expression, Product Of Sum (POS) expression - min term and max term - simplification of Boolean expressions using logic rules - truth tables - Karnaugh map - 2, 3, 4 variables - K-map reduction - don't cares in K- map-advantages and disadvantages of K- Map- exercise problems Design the Combinational circuits such as full adder, full subtractor, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Multiplexer, Gray code converter

This module is presented from the official syllabus scope for Digital Circuits and Systems. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Multiplexer, Gray code converter
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ Design the full adder circuit using universal gates 2 Applying Design the full subtractor circuit using universal

Design the full adder circuit using universal gates 2 Applying Design the full subtractor circuit using universal is an official topic in Module 2 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Design the full adder circuit using universal gates 2 Applying Design the full subtractor circuit using universal using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, design the full adder circuit using universal gates 2 applying design the full subtractor circuit using universal should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Design the full adder circuit using universal gates 2 Applying Design the full subtractor circuit using universal means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Design the full adder circuit using universal gates 2 Applying Design the full subtractor circuit using universal □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ 2 Applying gates Design the multiplexer and de multiplexer circuit

2 Applying gates Design the multiplexer and de multiplexer circuit is an official topic in Module 2 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying gates Design the multiplexer and de multiplexer circuit using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying gates design the multiplexer and de multiplexer circuit should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 2 Applying gates Design the multiplexer and de multiplexer circuit means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Applying gates Design the multiplexer and de multiplexer circuit □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 3 Applying using universal gates

3 Applying using universal gates is an official topic in Module 2 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying using universal gates using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying using universal gates should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying using universal gates means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 2-□□ 3 Applying using universal gates □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Design the BCD to decimal decoder 2 Applying Design the Gray to Binary code converter using

Design the BCD to decimal decoder 2 Applying Design the Gray to Binary code converter using is an official topic in Module 2 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Design the BCD to decimal decoder 2 Applying Design the Gray to Binary code converter using using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, design the bcd to decimal decoder 2 applying design the gray to binary code converter using should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Design the BCD to decimal decoder 2 Applying Design the Gray to Binary code converter using means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Design the BCD to decimal decoder 2 Applying Design the Gray to Binary code converter using □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 2 Applying universal gates Design the Binary to Gray code converter using

2 Applying universal gates Design the Binary to Gray code converter using is an official topic in Module 2 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying universal gates Design the Binary to Gray code converter using using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying universal gates design the binary to gray code converter using should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 2 Applying universal gates Design the Binary to Gray code converter using means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Applying universal gates Design the Binary to Gray code converter using □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 2 Applying universal gates Combinational logic circuits - introduction - design half adder, full adder, half sub tractor and full sub tractor - parallel adder-multiplexer / data selector - 4 to 1 MUX - applications of MUX – De multiplexer - 1 to 4 De multiplexer - 3 bit encoder - decoders - BCD to decimal, Binary to Gray code and Gray to Binary code converter - exercise problems Design the Sequential logic circuits such as Asynchronous and

2 Applying universal gates Combinational logic circuits - introduction - design half adder, full adder, half sub tractor and full sub tractor - parallel adder-multiplexer / data selector - 4 to 1 MUX - applications of MUX – De multiplexer - 1 to 4 De multiplexer - 3 bit encoder - decoders - BCD to decimal, Binary to Gray code and Gray to Binary code converter - exercise problems Design the Sequential logic circuits such as Asynchronous and is an official topic in Module 2 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

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Study points

- Define or identify 2 Applying universal gates Combinational logic circuits - introduction - design half adder, full adder, half sub tractor and full sub tractor - parallel adder-multiplexer / data selector - 4 to 1 MUX - applications of MUX – De multiplexer - 1 to 4 De multiplexer - 3 bit encoder - decoders - BCD to decimal, Binary to Gray code and Gray to Binary code converter - exercise problems Design the Sequential logic circuits such as Asynchronous and using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying universal gates combinational logic circuits - introduction - design half adder, full adder, half sub tractor and full sub tractor - parallel adder-multiplexer / data selector - 4 to 1 mux - applications of mux – de multiplexer - 1 to 4 de multiplexer - 3 bit encoder - decoders - bcd to decimal, binary to gray code and gray to binary code converter - exercise problems design the sequential logic circuits such as asynchronous and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying universal gates Combinational logic circuits - introduction - design half adder, full adder, half sub tractor and full sub tractor - parallel adder-multiplexer / data selector - 4 to 1 MUX - applications of MUX – De multiplexer - 1 to 4 De multiplexer - 3 bit encoder - decoders - BCD to decimal, Binary to Gray code and Gray to Binary code converter - exercise problems Design the Sequential logic circuits such as Asynchronous and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Applying universal gates Combinational logic circuits - introduction - design half adder, full adder, half sub tractor and full sub tractor - parallel adder-multiplexer / data selector - 4 to 1 MUX - applications of MUX – De multiplexer - 1 to 4 De multiplexer - 3 bit encoder - decoders - BCD to decimal, Binary to Gray code and Gray to Binary code converter - exercise problems Design the Sequential logic circuits such as Asynchronous and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Synchronous counter

This module is presented from the official syllabus scope for Digital Circuits and Systems. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Synchronous counter
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

▼ 1 Understanding asynchronous sequential logic circuits Explain SR, JK , T D, flip-flop using NAND with

1 Understanding asynchronous sequential logic circuits Explain SR, JK , T D, flip-flop using NAND with is an official topic in Module 3 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding asynchronous sequential logic circuits Explain SR, JK , T D, flip-flop using NAND with using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding asynchronous sequential logic circuits explain sr, jk , t d, flip-flop using nand with should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 1 Understanding asynchronous sequential logic circuits Explain SR, JK , T D, flip-flop using NAND with means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 1 Understanding asynchronous sequential logic circuits Explain SR, JK , T D, flip-flop using NAND with □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 2 Understanding the help of truth table Explain the working of master slave JK flip flop

2 Understanding the help of truth table Explain the working of master slave JK flip flop is an official topic in Module 3 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding the help of truth table Explain the working of master slave JK flip flop using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding the help of truth table explain the working of master slave jk flip flop should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 2 Understanding the help of truth table Explain the working of master slave JK flip flop means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding the help of truth table Explain the working of master slave JK flip flop □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 2 Understanding using NAND gates Explain the working of shift registers:-serial-in serial-out, parallel-in parallel-out, parallel-in

2 Understanding using NAND gates Explain the working of shift registers:-serial-in serial-out, parallel-in parallel-out, parallel-in is an official topic in Module 3 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding using NAND gates Explain the working of shift registers:-serial-in serial-out, parallel-in parallel-out, parallel-in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding using nand gates explain the working of shift registers:-serial-in serial-out, parallel-in parallel-out, parallel-in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding using NAND gates Explain the working of shift registers:-serial-in serial-out, parallel-in parallel-out, parallel-in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding using NAND gates Explain the working of shift registers:-serial-in serial-out, parallel-in parallel-out, parallel-in □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 2 Understanding serial-out and serial- in parallel-out using flip flops Explain the working of ring counter, Johnson

2 Understanding serial-out and serial- in parallel-out using flip flops Explain the working of ring counter, Johnson is an official topic in Module 3 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding serial-out and serial- in parallel-out using flip flops Explain the working of ring counter, Johnson using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding serial-out and serial- in parallel-out using flip flops explain the working of ring counter, johnson should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding serial-out and serial- in parallel-out using flip flops Explain the working of ring counter, Johnson means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding serial-out and serial- in parallel-out using flip flops Explain the working of ring counter, Johnson □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 2 Understanding counter using flip flops

2 Understanding counter using flip flops is an official topic in Module 3 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding counter using flip flops using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding counter using flip flops should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding counter using flip flops means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding counter using flip flops □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

▼ Design asynchronous mod-N counter

Design asynchronous mod-N counter is an official topic in Module 3 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Design asynchronous mod-N counter using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, design asynchronous mod-n counter should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Design asynchronous mod-N counter means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Design asynchronous mod-N counter □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

▼ Design mod-8 synchronous counter

Design mod-8 synchronous counter is an official topic in Module 3 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Design mod-8 synchronous counter using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, design mod-8 synchronous counter should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Design mod-8 synchronous counter means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Design mod-8 synchronous counter □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ Design 3-bit up down counter 2 Applying Sequential logic circuits - introduction - synchronous and asynchronous sequential logic circuits - SR flip flop - SR latch - SR flip flop using NAND & NOR gates - JK flip flop with preset and clear inputs - D flip flop - T flip flop - master slave JK flip-flop - shift registers - serial in serial out, parallel in parallel out, serial in parallel out, parallel in serial out shift registers - left shift and right shift registers - applications of shift registers - ring counter - Johnson counter - Binary counters - design asynchronous mod-N counter ,mod-8 synchronous counter, 3-bit up down counter

Design 3-bit up down counter 2 Applying Sequential logic circuits - introduction - synchronous and asynchronous sequential logic circuits - SR flip flop - SR latch - SR flip flop using NAND & NOR gates - JK flip flop with preset and clear inputs - D flip flop - T flip flop - master slave JK flip-flop - shift registers - serial in serial out, parallel in parallel out, serial in parallel out, parallel in serial out shift registers - left shift and right shift registers - applications of shift registers - ring counter - Johnson counter - Binary counters - design asynchronous mod-N counter ,mod-8 synchronous counter, 3-bit up down counter is an official topic in Module 3 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Design 3-bit up down counter 2 Applying Sequential logic circuits - introduction - synchronous and asynchronous sequential logic circuits - SR flip flop - SR latch - SR flip flop using NAND & NOR gates - JK flip flop with preset and clear inputs - D flip flop - T flip flop - master slave JK flip-flop - shift registers - serial in serial out, parallel in parallel out, serial in parallel out, parallel in serial out shift registers - left shift and right shift registers - applications of shift registers - ring counter - Johnson counter - Binary counters - design asynchronous mod-N counter ,mod-8 synchronous counter, 3-bit up down counter using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, design 3-bit up down counter 2 applying sequential logic circuits - introduction - synchronous and asynchronous sequential logic circuits - sr flip flop - sr latch - sr flip flop using nand & nor gates - jk flip flop with preset and clear inputs - d flip flop - t flip flop - master slave jk flip-flop - shift registers - serial in serial out, parallel in parallel out, serial in parallel out, parallel in serial out shift registers - left shift and right shift registers - applications of shift registers - ring counter - johnson counter - binary counters - design asynchronous mod-n counter ,mod-8 synchronous counter, 3-bit up down counter should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Design 3-bit up down counter 2 Applying Sequential logic circuits - introduction - synchronous and asynchronous sequential logic circuits - SR flip flop - SR latch - SR flip flop using NAND & NOR gates - JK flip flop with preset and clear inputs - D flip flop - T flip flop - master slave JK flip-flop - shift registers - serial in serial out, parallel in parallel out, serial in parallel out, parallel in serial out shift registers - left shift and right shift registers - applications of shift registers - ring counter - Johnson counter - Binary counters - design asynchronous mod-N counter ,mod-8

Aspect	What to prepare
	synchronous counter, 3-bit up down counter means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Design 3-bit up down counter 2 Applying Sequential logic circuits - introduction - synchronous and asynchronous sequential logic circuits - SR flip flop - SR latch - SR flip flop using NAND & NOR gates - JK flip flop with preset and clear inputs - D flip flop - T flip flop - master slave JK flip-flop - shift registers - serial in serial out, parallel in parallel out, serial in parallel out, parallel in serial out shift registers - left shift and right shift registers - applications of shift registers - ring counter - Johnson counter - Binary counters - design asynchronous mod-N counter ,mod-8 synchronous counter, 3-bit up down counter □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the

Malayalam support notes.

MODULE 4

Describe the working of ADC and DAC

This module is presented from the official syllabus scope for Digital Circuits and Systems. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the working of ADC and DAC
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 1 Understanding DAC

1 Understanding DAC is an official topic in Module 4 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding DAC using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding dac should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding DAC means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 1 Understanding DAC □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Describe the working of R-2R ladder type DAC 1 Understanding Describe the working of various Analog to

Describe the working of R-2R ladder type DAC 1 Understanding Describe the working of various Analog to is an official topic in Module 4 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the working of R-2R ladder type DAC 1 Understanding Describe the working of various Analog to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the working of r-2r ladder type dac 1 understanding describe the working of various analog to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the working of R-2R ladder type DAC 1 Understanding Describe the working of various Analog to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Describe the working of R-2R ladder type DAC 1 Understanding Describe the working of various Analog to □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 2 Understanding Digital Converter

2 Understanding Digital Converter is an official topic in Module 4 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Digital Converter using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding digital converter should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Digital Converter means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 2 Understanding Digital Converter □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ **List the specifications of DAC and ADC 1 Understanding DAC - specifications - resolution, accuracy, settling time - different types - binary weighted resister method and R-2R ladder type - ADC - counter type - successive approximation type Text / Reference: T/R Book Title/Author Digital fundamentals – Thomas L. Floyd, PEARSON EDUCATION publication, R1 Eleventh edition – Global Edition, Digital Electronics –principles and integrated circuits. Anil K. Maini. Wiley R2 publications, first edition. R3 Fundamentals of Digital circuits, Anandkumar. A ,PHI Digital principles and applications. Donald P Leach, Albert Paul Malvino, R4 Goutam Saha, McGraw Hill Publisher, 7th edition Digital Computer Fundamentals,-Thomas C Bartee, McGraw-Hill Publisher,4th R5 edition. Online Resources: Sl.No Website Link 1 https://www.digitalelectronicsdeeds.com/ 2 https://www.electrical4u.com/digital-electronics/ 3 https://www.tutorialspoint.com/digital_circuits/index.htm**

List the specifications of DAC and ADC 1 Understanding DAC - specifications - resolution, accuracy, settling time - different types - binary weighted resister method and R-2R ladder type - ADC - counter type - successive approximation type Text / Reference: T/R Book Title/Author Digital fundamentals – Thomas L. Floyd, PEARSON EDUCATION publication, R1 Eleventh edition – Global Edition, Digital Electronics –principles and integrated circuits. Anil K. Maini. Wiley R2 publications, first edition. R3 Fundamentals of Digital circuits, Anandkumar. A ,PHI Digital principles and applications. Donald P Leach, Albert Paul Malvino, R4 Goutam Saha, McGraw Hill Publisher, 7th edition Digital Computer Fundamentals,-Thomas C Bartee, McGraw-Hill Publisher,4th R5 edition. Online Resources: Sl.No Website Link 1 https://www.digitalelectronicsdeeds.com/ 2 https://www.electrical4u.com/digital-electronics/ 3 https://www.tutorialspoint.com/digital_circuits/index.htm is an official topic in Module 4 of Digital Circuits and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List the specifications of DAC and ADC 1 Understanding DAC - specifications - resolution, accuracy, settling time - different types - binary weighted resister method and R-2R ladder type - ADC - counter type - successive approximation type Text / Reference: T/R Book Title/Author Digital fundamentals – Thomas L. Floyd, PEARSON EDUCATION publication, R1 Eleventh edition – Global Edition, Digital Electronics –principles and integrated circuits. Anil K. Maini. Wiley R2 publications, first edition. R3 Fundamentals of Digital circuits, Anandkumar. A ,PHI Digital principles and applications. Donald P Leach, Albert Paul Malvino, R4 Goutam Saha, McGraw Hill Publisher, 7th edition Digital Computer Fundamentals,-Thomas C Bartee, McGraw-Hill Publisher,4th R5 edition. Online Resources: Sl.No Website Link 1 https://www.digitalelectronicsdeeds.com/ 2 https://www.electrical4u.com/digital-electronics/ 3 https://www.tutorialspoint.com/digital_circuits/index.htm using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list the specifications of dac and adc 1 understanding dac - specifications - resolution, accuracy, settling time - different types - binary weighted resister method and r-2r ladder type - adc - counter type - successive approximation type text / reference: t/r book title/author digital fundamentals – thomas l. floyd, pearson education publication, r1 eleventh edition – global edition, digital electronics –principles and integrated circuits. anil k. maini. wiley r2 publications, first edition. r3 fundamentals of digital circuits, anandkumar. a ,phi digital principles and applications. donald p leach, albert paul malvino, r4 goutam saha, mcgraw hill publisher, 7th edition digital computer fundamentals,-thomas c bartee, mcgraw-hill publisher,4th r5 edition. online resources: sl.no website link 1 https://www.digitalelectronicsdeeds.com/ 2 https://www.electrical4u.com/digital-electronics/ 3 https://www.tutorialspoint.com/digital_circuits/index.htm should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What List the specifications of DAC and ADC 1 Understanding DAC - specifications - resolution, accuracy, settling time - different types - binary weighted resister method and R-2R ladder type - ADC - counter type - successive approximation type Text / Reference: T/R Book Title/Author Digital fundamentals – Thomas L. Floyd, PEARSON EDUCATION publication, R1 Eleventh edition – Global Edition, Digital Electronics –principles and integrated circuits. Anil K. Maini. Wiley R2 publications, first edition. R3 Fundamentals of Digital circuits, Anandkumar. A ,PHI Digital principles and applications. Donald P Leach, Albert Paul Malvino, R4 Goutam Saha, McGraw Hill Publisher, 7th edition Digital Computer Fundamentals,-Thomas C Bartee, McGraw-Hill Publisher,4th R5 edition. Online Resources: Sl.No Website Link 1 https://www.digitalelectronicsdeeds.com/ 2 https://www.electrical4u.com/digital-electronics/ 3 https://www.tutorialspoint.com/digital_circuits/index.htm means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 4-□□ List the specifications of DAC and ADC 1 Understanding DAC - specifications - resolution, accuracy, settling time - different types - binary weighted resister method and R-2R ladder type - ADC - counter type - successive approximation type Text / Reference: T/R Book Title/Author Digital fundamentals – Thomas L. Floyd, PEARSON EDUCATION publication, R1 Eleventh edition – Global Edition, Digital Electronics –principles and integrated circuits. Anil K. Maini. Wiley R2 publications, first edition. R3 Fundamentals of Digital circuits, Anandkumar. A ,PHI Digital principles and applications. Donald P Leach, Albert Paul Malvino, R4 Goutam Saha, McGraw Hill Publisher, 7th edition Digital Computer Fundamentals,-Thomas C Bartee, McGraw-Hill Publisher,4th R5 edition. Online Resources: Sl.No Website Link 1 https://www.digitalelectronicsdeeds.com/ 2 https://www.electrical4u.com/digital-electronics/ 3 https://www.tutorialspoint.com/digital_circuits/index.htm □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain 4 Applying operations on different number systems. (3 marks)**

▶ **Define or explain Build logical expressions using logic gates 3 Applying Simplify the Boolean expressions using K map and. (3 marks)**

▶ **Define or explain 4 Applying logic rules Number systems - Decimal, Binary and Hexa Decimal number systems - conversion - use of binary codes - types of binary codes - Binary Coded Decimal, Excess 3 code, Gray code, ASCII code Binary addition, subtraction, multiplication and division - 1’s complement and 2’s complement subtraction - introduction to logic gates - AND, OR, NOT, NAND, NOR, EX-OR and EX-NOR operations - universal property of NAND and NOR gates - realization of AND, OR, NOT, EX-OR and EX-NOR -Laws of Boolean algebra and De- Morgan's theorems - Sum Of Products (SOP) expression, Product Of Sum (POS) expression - min term and max term - simplification of Boolean expressions using logic rules - truth tables - Karnaugh map - 2, 3, 4 variables - K-map reduction - don't cares in K-map-advantages and disadvantages of K- Map- exercise problems Design the Combinational circuits such as full adder, full subtractor,. (3 marks)**

Module 2

▶ **Define or explain Design the full adder circuit using universal gates 2**

Applying Design the full subtractor circuit using universal. (3 marks)

▶ **Define or explain 2 Applying gates Design the multiplexer and de multiplexer circuit. (3 marks)**

▶ **Define or explain 3 Applying using universal gates. (3 marks)**

▶ **Define or explain Design the BCD to decimal decoder 2 Applying Design the Gray to Binary code converter using. (3 marks)**

Module 3

▶ **Define or explain 1 Understanding asynchronous sequential logic circuits Explain SR, JK , T D, flip-flop using NAND with. (3 marks)**

▶ **Define or explain 2 Understanding the help of truth table Explain the working of master slave JK flip flop. (3 marks)**

▶ **Define or explain 2 Understanding using NAND gates Explain the working of shift registers:-serial-in serial-out, parallel-in parallel-out, parallel-in. (3 marks)**

▶ **Define or explain 2 Understanding serial-out and serial- in parallel-out using flip flops Explain the working of ring counter, Johnson. (3 marks)**

Module 4

▶ **Define or explain 1 Understanding DAC. (3 marks)**

▶ **Define or explain Describe the working of R-2R ladder type DAC 1 Understanding Describe the working of various Analog to. (3 marks)**

▶ **Define or explain 2 Understanding Digital Converter. (3 marks)**

▶ **Define or explain List the specifications of DAC and ADC 1 Understanding DAC - specifications - resolution, accuracy, settling time - different types - binary weighted resister method and R-2R ladder type - ADC - counter type - successive approximation type Text / Reference: T/R Book Title/Author Digital fundamentals – Thomas L. Floyd, PEARSON EDUCATION publication, R1 Eleventh edition – Global Edition, Digital Electronics –principles and integrated circuits. Anil K. Maini. Wiley R2 publications, first edition. R3 Fundamentals of Digital circuits, Anandkumar. A ,PHI Digital principles and applications. Donald P Leach, Albert Paul Malvino, R4 Goutam Saha, McGraw Hill Publisher, 7th edition Digital Computer Fundamentals,-Thomas C Bartee, McGraw-Hill Publisher,4th R5 edition. Online Resources: Sl.No Website Link 1 https://www.digitalelectronicsdeeds.com/ 2 https://www.electrical4u.com/digital-electronics/ 3 https://www.tutorialspoint.com/digital_circuits/index.htm. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **4 Applying operations on different number systems**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Design the full adder circuit using universal gates 2 Applying Design the full subtractor circuit using universal**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **1 Understanding asynchronous sequential logic circuits Explain SR, JK , T D, flip-flop using NAND with**.

Module 4

1-mark definition: State the meaning of **1 Understanding DAC**.

3-mark explanation: Explain one principle, process or comparison from this module.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://www.digitalelectronicsdeeds.com/
- https://www.electrical4u.com/digital-electronics/
- https://www.tutorialspoint.com/digital_circuits/index.htm

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 3081. Verify institutional updates against the current official curriculum.